

GENERAL PHYSICS LABORATORY II
Department of Physics, University of Colombo.

Final Group Project

A Cost-Effective Polarimeter for Determining the Concentration and Chirality of Glucose.

Name : K.D.S.S.N.Hemachandra
Registration No : 2023s20018
Index No : 17244
Group No : 06

1. ABSTRACT

This study presents the design and implementation of a cost-effective polarimeter for determining the concentration and chirality of glucose solutions. The device utilizes a parabolic reflector-based LED light source, polarizing filters, a stepper-motor controlled analyzer, and an RGB light sensor to measure optical rotation. A set of glucose solutions with varying concentrations was analyzed to investigate the relationship between optical rotation and concentration. The results demonstrate a linear correlation between concentration and optical rotation, consistent with Biot's law of optical rotation. The experimental specific rotation of glucose was determined to be **53.1 ± 4.8 degrees per ml g⁻¹ dm⁻¹ at 27°C and 620 nm wavelength**, which deviates by 0.76% from the theoretical value of 52.70 degrees per ml g⁻¹ dm⁻¹ at 25°C. The observed clockwise rotation confirmed the chirality of the glucose sample as D-glucose. The study highlights the effectiveness of a low-cost approach to polarimetry, with potential applications in educational and industrial settings for optically active substance analysis.

2. INTRODUCTION

Light is an electromagnetic wave that oscillates in all directions perpendicular to its direction of propagation. However, when light interacts with certain materials, it can become polarized, meaning its oscillations are confined to a single plane. Polarization plays a crucial role in various optical applications, including imaging, display technologies, and analytical chemistry. One of the primary methods for achieving polarization is through polarizing filters, which selectively transmit light waves oscillating in a particular direction while blocking others. This principle is fundamental in polarimetry, a technique used to study optically active substances.

Optically active substances, such as glucose, rotate the plane of polarization of light due to their chiral molecular structures. Chirality arises when a molecule exists in two non-superimposable mirror image forms, commonly labeled as **D** (*Dextrorotatory*) and **L** (*Levorotatory*) isomers. This rotation of polarized light is governed by Biot's law of optical rotation, which states that the angle of rotation (θ) is directly proportional to the concentration (c) of the optically active substance, the path length (l) of light through the sample, and the substance's specific rotation ($[\alpha]$), expressed as:

$$\theta = [\alpha]_T^{\lambda} \cdot c \cdot l \text{ ----- (1)}$$

Several factors influence optical rotation, including temperature, the wavelength of light, and the concentration of the solution. Understanding these factors is essential for accurate polarimetric measurements.

When polarized light passes through another polarizer, the transmitted intensity follows Malus' Law, which states that the intensity (I) after passing through a second polarizer is given by:

$$I = I_0 \cos^2\theta \text{ -----(2)}$$

where I_0 is the initial intensity, and θ is the angle between the axes of the two polarizers. In a polarimeter, this principle is used to measure the angle of rotation by determining the change in intensity as the analyzer (second polarizer) is rotated.

Polarimetry has significant applications in both industry and research. In the sugar industry, it is used for quality control by determining the sugar concentration in solutions. In pharmaceuticals, precise identification of chiral molecules is crucial, as incorrect enantiomers can lead to adverse drug effects. Additionally, in clinical diagnostics, polarimetry aids in measuring glucose levels in biological samples, which is essential for diabetes management. However, commercial polarimeters are often expensive, limiting their accessibility in resource constrained settings.

This study aims to design and construct a cost-effective, portable polarimeter capable of determining the chirality and concentration of glucose solutions. By investigating the relationship between the angle of optical rotation and concentration, the specific rotation of glucose can be determined, establishing a practical method for analyzing optically active solutions. The primary research problem addressed in this project is how to measure the chirality and specific rotation of glucose using a simple yet accurate polarimetric technique. Since this is an open-ended project, its applications can extend further. If properly calibrated with known concentration samples, the device can be used to determine the concentration of any optically active substance. This contributes to the ongoing effort to develop accessible, low cost scientific instruments for both educational and industrial applications.

3. METHODOLOGY

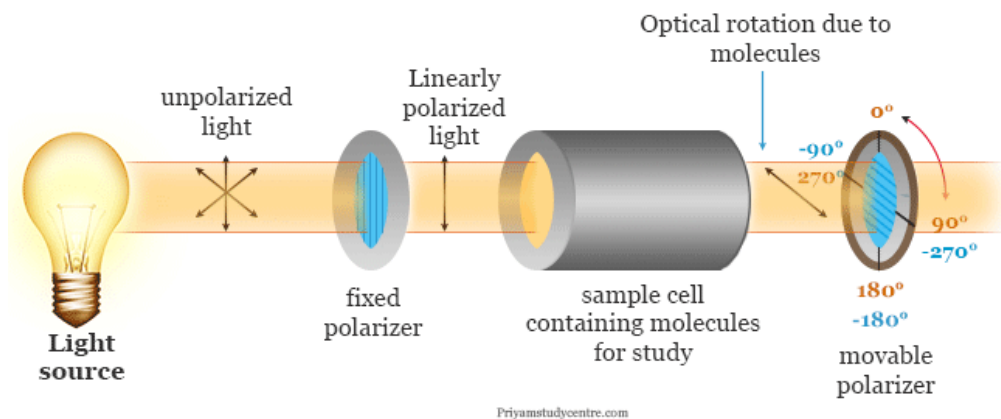


Figure 3.1: Polarimetry instrumentation. Unpolarized light passes through a fixed polarizer, producing linearly polarized light. The light then travels through the optically active substance and the analyzer. The optical rotation angle is determined by rotating the analyzer.

3.1. Materials and Apparatus

The experimental setup included **two polarizing filters**, extracted from an LCD screen, and an **SMD LED** emitting RGB colors, which was soldered into a parabolic reflector. An **RGB light sensor** (TCS3472) with a photo-diode was used to detect light intensity. The analyzer was rotated using a **stepper motor** (28BJ-48) driven by a ULN2003 driver module. A **glass cuvette** (16 cm × 4 cm × 3 cm) was used as the solution holder. The **sample holder** and **two**

spur gears (25-tooth and 100-tooth) were fabricated using a 3D printer. A **breadboard**, **jumper cables**, and **connecting wires** were used to assemble the circuit.

3.2.Preparation and placement of Glucose Solutions.

A **3.5M glucose stock solution** was prepared by dissolving a measured mass of glucose in distilled water (*Figure 3.2.c*). The required glucose mass was measured using an electronic balance, and the water volume was determined with graduated cylinders of various capacities. The solution was stirred until complete dissolution was achieved.

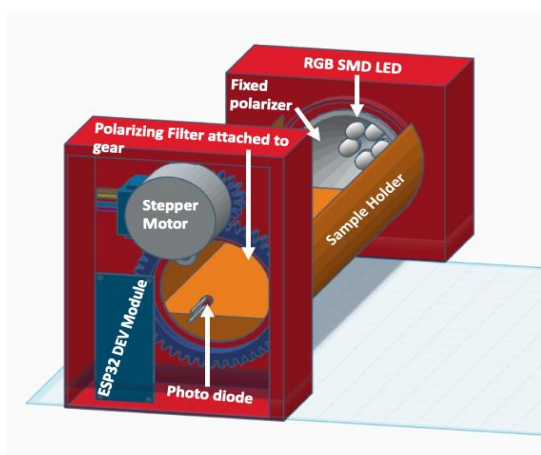
To obtain different concentrations, systematic dilution was performed in 0.50M increments, resulting in six glucose solutions of varying concentrations. Each solution was mixed thoroughly to maintain uniformity.

Before measuring optical rotation, a **blank solution** (normal water) was used to establish the zero-reference position. This ensured that all subsequent optical rotation values were measured relative to this calibrated baseline, reducing systematic errors. For measurement, each glucose solution was carefully poured into a clean, dry cuvette and placed inside the polarimeter (*Figure 3.2.d*). After each trial, the cuvette was thoroughly washed with water and dried completely to prevent cross-contamination between measurements.

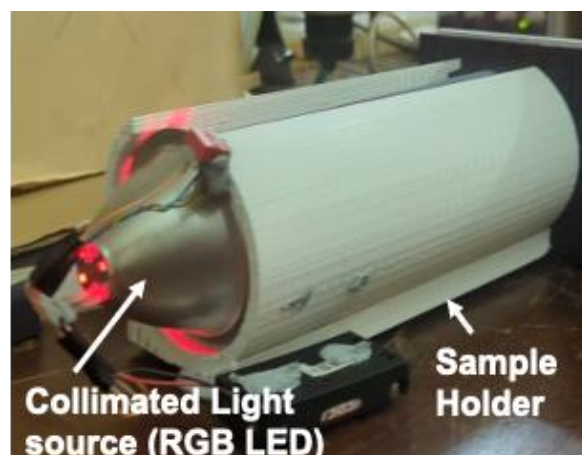
3.3.Polarimeter Measurement and analysis

The polarimeter was aligned to ensure proper light transmission through the polarizer, glucose sample, and analyzer. The analyzer was then rotated using a stepper motor to identify the position of minimum light intensity (dark point), corresponding to the angle of optical rotation. The intensity variations were detected using an RGB light sensor and a photo-diode, enabling precise measurements. This process was repeated for all glucose concentrations to obtain accurate rotation values.

The recorded angles were used to calculate the specific rotation of glucose using equation (1). A **graph of optical rotation vs. concentration** was plotted to analyze the relationship between these variables. The chirality of the glucose sample was determined based on the direction of optical rotation.



(a)



(b)

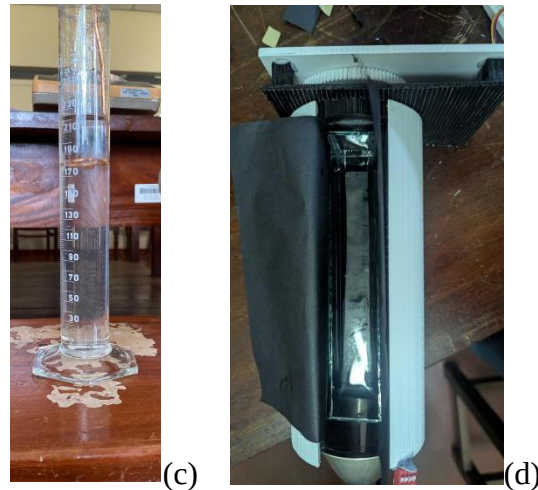


Figure 3.2: (a) 3D model of the polarimeter apparatus. (b) Actual assembled polarimeter setup with a parabolic reflector and a 3D-printed sample holder. (c) Well-dissolved glucose stock solution with a 3.5M concentration. (d) Top view of the apparatus with the glass cuvette placed inside.

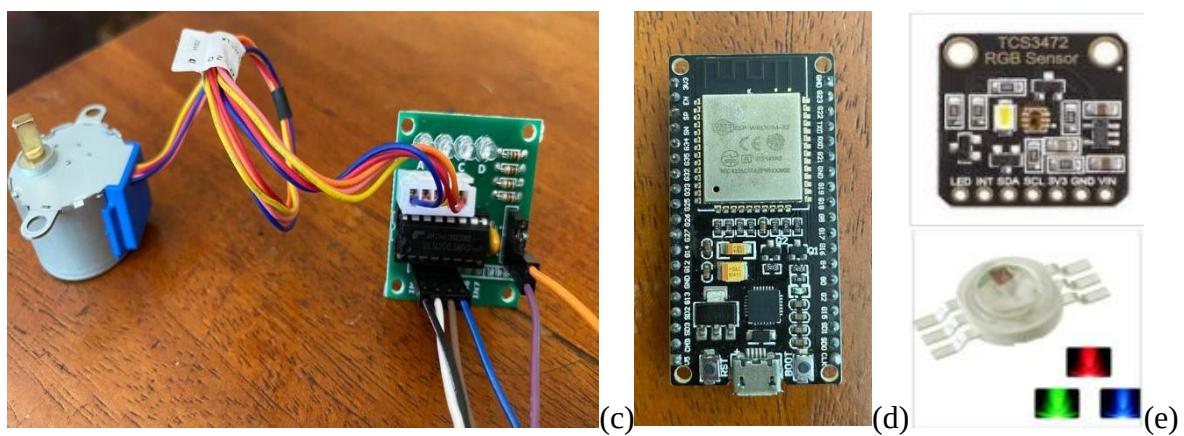
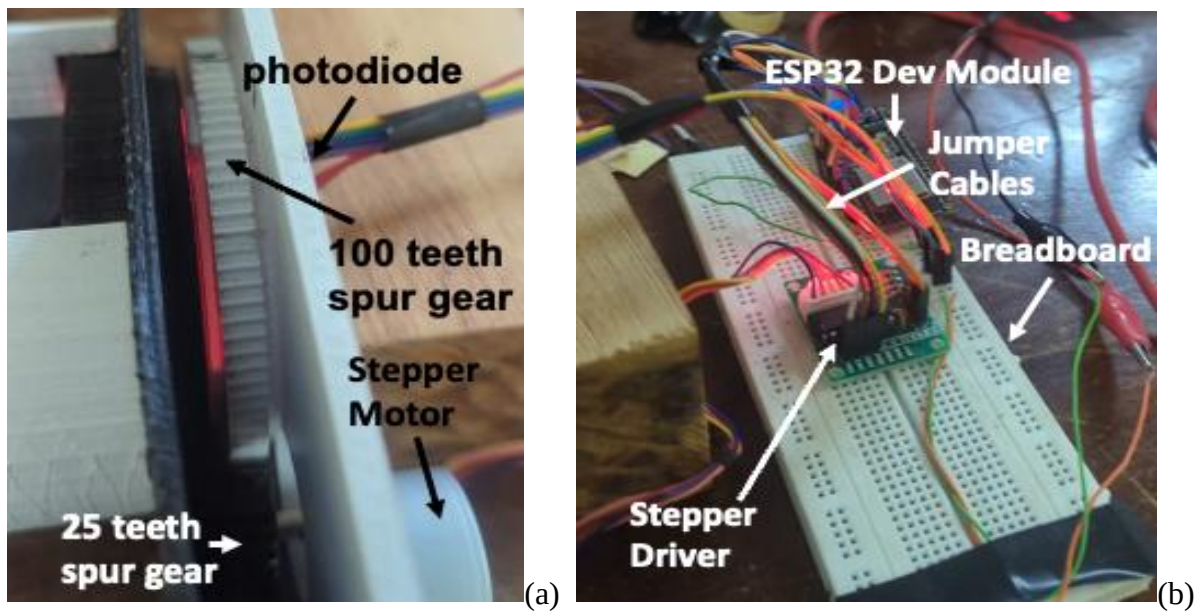


Figure 3.3: (a) Zoomed view of the analyzing section, showing the 100-tooth larger gear and 25-tooth smaller gear rotated by the stepper motor. (b) Stepper driver used to control the stepper motor, with the ESP32 connected to a breadboard using jumper cables. (c) Stepper motor (28BJ-48) and driver (ULN2003)

responsible for rotating and controlling the analyzer. (d) ESP32 micro-controller used for data processing, control, and connectivity in the polarimeter setup. (e) TCS3472 RGB light sensor and SMD RGB LED. The sensor detects the intensity of transmitted light after passing through the sample, while the SMD RGB LED emits red, green, and blue light with 1W power for illumination.

4. RESULTS and DISCUSSION

Minimum reading of the stepper motor : 0.5°

Electronic balance minimum reading: 0.01g

Table 1: Recorded data using the experimental setup

Concentration (mol dm ⁻³)	Concentration (g/cm ³)	Optical rotation (θ) ($\pm 0.5^{\circ}$)
3.50	0.630	45.4°
3.25	0.585	45.7°
2.75	0.495	34.3°
2.25	0.405	32.4°
1.75	0.315	22.0°
1.25	0.225	18.4°

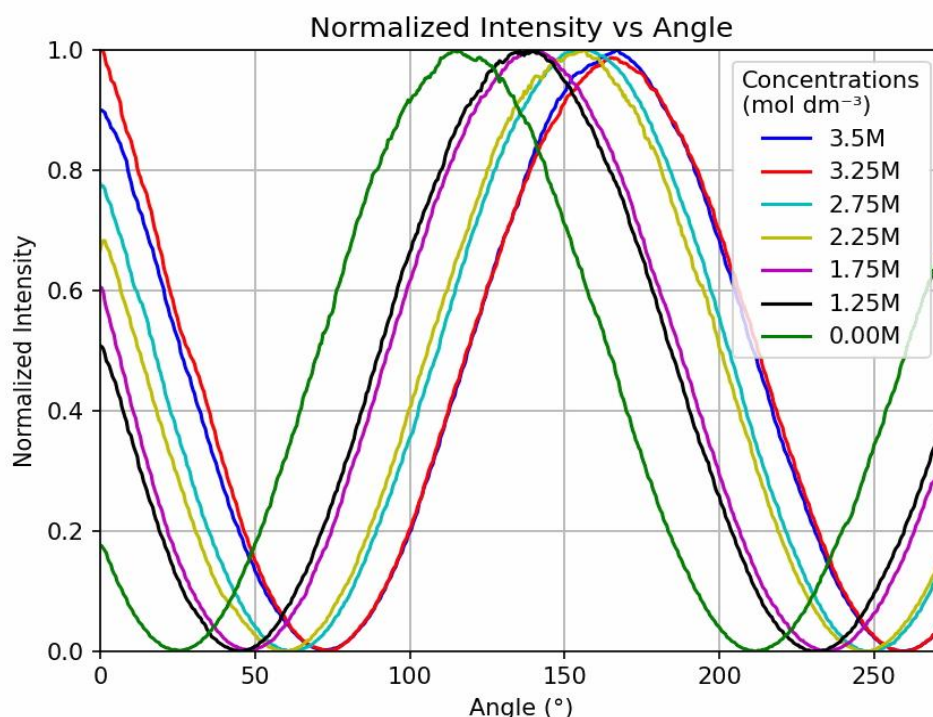


Figure 4.1: Normalized Intensity vs. Angle (Malus' Law Proof)- this graph demonstrates Malus' Law by plotting the normalized intensity (I_0) as a function of the analyzer angle (θ). The curves correspond to different glucose concentrations (in mol dm⁻³), showing how the intensity follows the expected $\cos^2\theta$ dependence. The phase shift between curves increases with concentration, indicating optical rotation due to the chiral nature of glucose.

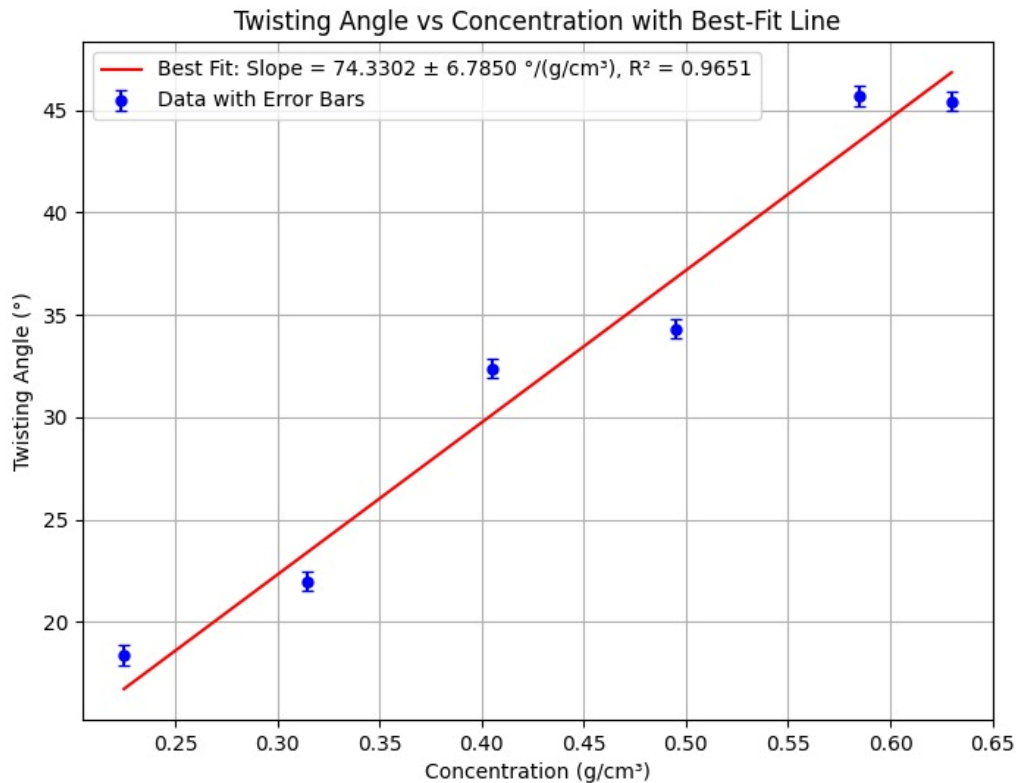


Figure 4.2: Average Minimum Angle Difference vs. Concentration -this graph presents the relationship between glucose concentration (in g/cm^3) and the average minimum angle difference observed due to optical rotation. The error bars represent the instrumental uncertainty, determined by the minimum readable step of the stepper motor controlling the analyzer. The best-fit linear regression, showing a high correlation ($R^2=0.9651$), with the gradient ($m=74.3302\text{ }^\circ/(\text{g/cm}^3)^{-1}$) corresponding to the product of the specific rotation and the path length (16 cm).

Gradient of the angle vs concentration graph: $74.3 \pm 6.8\text{ }^\circ/(\text{g/cm}^3)^{-1}$

- Experimental value of specific rotation calculated using equation (1) :
 $53.1 \pm 4.8\text{ degrees per ml g}^{-1}\text{ dm}^{-1}$ (27°C and 620nm wavelength)
- Expected value of specific rotation of glucose in 25°C :
 $52.7^\circ\text{ degrees per ml g}^{-1}\text{ dm}^{-1}$

$$\text{Percentage deviation from the actual value} = \frac{|52.7-53.1|}{52.7} \times 100\% = 0.76\%$$

$$\text{Error percentage} = \frac{4.8}{53.1} \times 100\% = 9.04\%$$

The results obtained from the polarimeter measurements indicate a clear relationship between glucose concentration and optical rotation. The measured angles of rotation for different glucose concentrations followed a consistent increasing trend, verifying Biot's Law, which states that optical rotation is directly proportional to concentration and path length. The recorded data showed that the highest concentration produced an higher optical rotation, while the lowest concentration resulted lower optical rotation. The experimental gradient of the optical rotation vs. concentration graph was determined to be $74.3 \pm 6.8\text{ }^\circ/(\text{g/cm}^3)^{-1}$, reflecting a strong linear correlation ($R^2 = 0.9651$) between the two variables. This confirms that the designed polarimeter accurately captures the optical activity of glucose.

Further verification of Malus' Law was achieved through the intensity vs. angle graph (*Figure 4.1*), which demonstrated the expected $\cos^2\theta$ dependence. The increasing phase shift in intensity with higher concentrations indicates the cumulative optical rotation effect of glucose. Additionally, the experimental determination of specific rotation resulted in a value of 53.1 ± 4.8 degrees per ml g⁻¹ dm⁻¹ at 27°C and a wavelength of 620 nm. This deviates from the standard value of 52.70 degrees per ml g⁻¹ dm⁻¹ at 25°C by 0.76%, which can be attributed to temperature variations, instrumental limitations, and possible impurities in the glucose solution. The stepper motor's minimum resolution (0.50°) and the electronic balance's accuracy (0.01 g) contributed to the measurement uncertainty. The calculated error percentage of 9.04% is a reasonable value for an experimental setup, considering the inherent limitations in precision and external factors affecting measurements.

The observed positive rotation direction of the polarized light confirms that the tested glucose sample is dextrorotatory (D-glucose), aligning with known optical characteristics of glucose. Using light of different wavelengths, such as red, green, and blue, can enhance measurement accuracy by accounting for wavelength-dependent variations in optical rotation. This approach allows for more precise determination of specific rotation and can help correct discrepancies arising from single wavelength measurements.

The ability of the constructed polarimeter to effectively measure chirality and concentration demonstrates its potential as an affordable and practical tool for laboratories and educational purposes. The findings support the feasibility of using a low cost setup for optical rotation measurements, confirming its applicability in scientific and industrial environments.

5. CONCLUSION

The successful development and testing of a cost-effective polarimeter confirm its capability to accurately measure the optical rotation of glucose solutions. The results validated Biot's Law, with a strong linear relationship between concentration and rotation angle. The measured specific rotation closely matched theoretical expectations, with a minor deviation attributed to experimental conditions. The study further demonstrated the ability of the polarimeter to distinguish chirality, identifying the glucose sample as dextrorotatory (D-glucose). Despite minor instrumental limitations, the overall accuracy and effectiveness of the device highlight its potential for extensive use in resource limited settings. Future improvements in light collimation and stepper motor precision could further enhance accuracy, expanding the applicability of this low-cost polarimeter in chemical and biochemical analysis.

6. REFERENCES

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7. APPENDIX

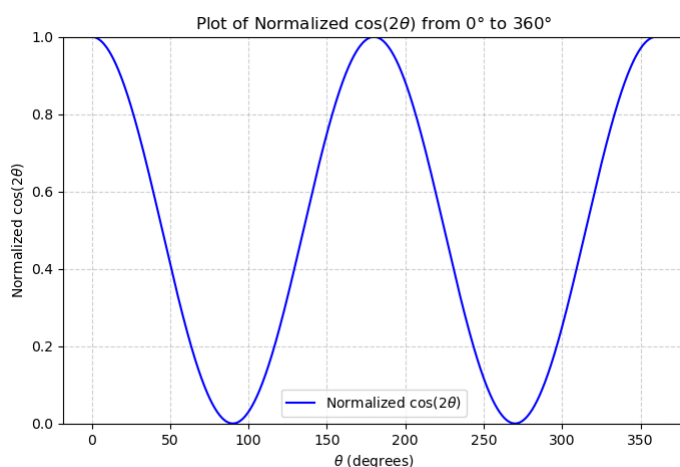
$$\theta = [\alpha_T^\lambda] \cdot l \cdot c$$

$$\text{Gradient} = [\alpha_T^\lambda] \cdot l = 74.3302^\circ/(\text{g}/\text{cm}^3)^{-1}$$

$$[\alpha_T^\lambda] = \frac{74.3302^\circ/(\text{g}/\text{cm}^3)^{-1}}{16\text{cm}} = 53.0930 \text{ degrees per ml g}^{-1} \text{ dm}^{-1} \sim 53.1$$

Slope (forced through origin): 74.3302 °/(g/cm³)
 R²: 0.9651
 Slope: 74.3302 ± 6.7850 °/(g/cm³)
 Specific Rotation: 53.0930 ± 4.8465 °·dm²/g
 R²: 0.9651

Specific Rotation at 27°C and 620 nm:
 $[\alpha]_{27^\circ 620} = 53.0930 \pm 4.8465^\circ \cdot \text{dm}^2/\text{g}$



The graph plotted to find the 0th position using blank solution(normal water). Then other values were taken relative to this position.